SHARIF UNIVERSITY OF TECHNOLOGY
Department of Computer Engineering

M.Sc. Thesis

Design of a Fault-Tolerant Deep Neural Network Hardware Accelerator for Image and Video Processing Applications

Presented by **Ali Roshani**
Supervisors Dr. M. Ansari & Dr. M. T. Manzuri Shalmani
Date Aban 1405

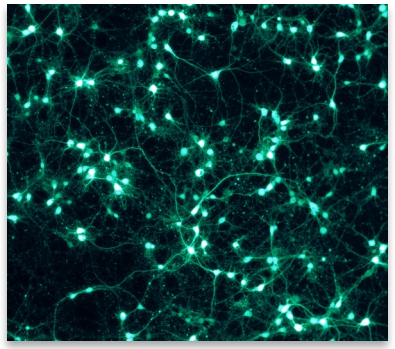
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Cyber-Physical systems Laboratory

Cyber-Physical Systems Laboratory



Outline

01 Introduction
02 Literature
03 Methodology
04 Results
05 Conclusion



01

Introduction

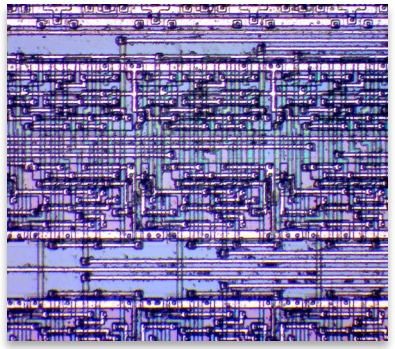
Mission-critical vision, the edge device, and why its neuron must not fail



02

Literature Review

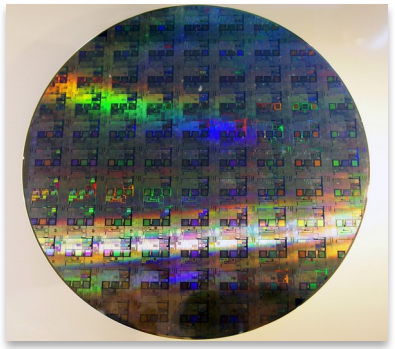
Six families of prior work and where this thesis stands



03

Methodology

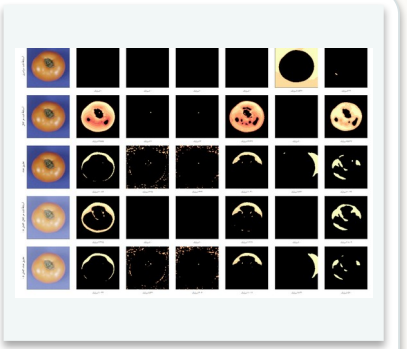
Hardened core → trustworthy fault analysis → BioVision



04

Results

PPA, fault campaign, RFoM, learning, BioVision



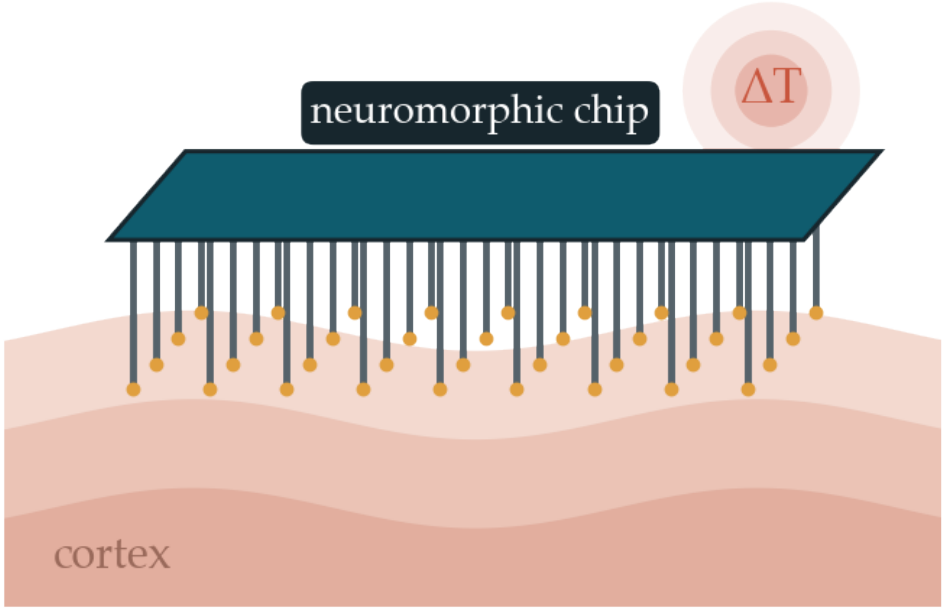
05

Conclusion

Takeaways, publications and the road ahead

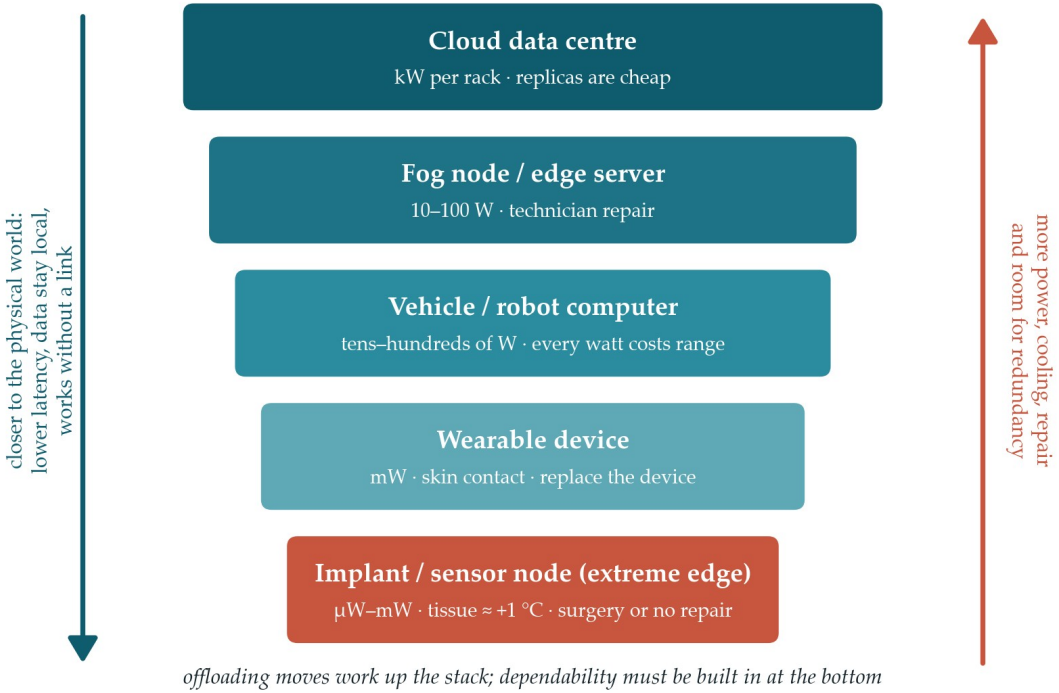
01

Introduction



Where Should Intelligence Live?

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1 Mission-critical vision

UAVs, autonomous cars, robots and retinal implants: the camera is the **first stage of a decision** that must be made in real time.

2 The systems answer: offloading

Cloudlets, fog and MEC move servers closer [54, 55, 56]; DNNs are split by layer [57]; split & federated learning [58, 59].

3 Its hidden assumption

The end device is a **black box that always computes correctly**; only its speed matters.

Down the ladder, power, heat headroom and repairability vanish — **an implant must stay within ≈ +1 °C**

[54] Satyanarayanan 2009 [55] Bonomi 2012 [56] Shi 2016 [57] Kang 2017 [58] Zhou 2019 [59] Kairouz 2021 [15] Kim 2007

Why Offloading Fails Safety-Critical Vision

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1 Time & determinism

Tail latency: 1 % slow × 100 servers → 63 % slow [61]; no worst-case bound; handover outages; model/state skew

2 Connectivity

Coverage gaps, jamming, radio loss in tissue and under water, cascading infrastructure failure

3 Energy & heat

1 kbit over 100 m ≈ 3 M instructions [62]; video saturates the uplink; radio heat and retransmissions

4 Security & privacy

Every hop is an attack surface; split learning leaks raw data [60]; trust in the operator

5 Certification

IEC 61508 / ISO 26262 need a bounded system [63]; unknown fault-containment boundary; liability

6 Silent corruption

Even cloud cores corrupt data silently [64]; the remote node is not a safe harbour

7 Scale & cost

Spectrum capacity, deployment and upkeep of the infrastructure

8 Environment & modelling

Radiation, heat and vibration hit the device anyway; offloading optimisers assume a fixed network model

24 weaknesses in 8 groups → **the critical perception–decision loop must close on the device itself**

[61] Dean & Barroso 2013 [62] Pottie & Kaiser 2000 [60] Pasquini 2021 [63] Koopman & Wagner 2016 [64] Hochschild 2021

Our Anchor: Build the Extreme-Edge Device

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A1 · Extreme edge

The safety-critical loop closes **on the device**; offloading stays for training, maps and analytics.

A2 · Purpose-built device

A device **designed for this task** — image and video — under size, weight, power, heat and cost limits.

Power envelope down the ladder

Platform	Typical power
CPU	10–100 W
GPU	≈ 100 W
FPGA	1–10 W
ASIC accelerator	0.1–1 W
Brain implant	**tens of mW (≈ +1 °C)**

3 Why spikes

- Brain: 10^{11} neurons at ≈ 20 W [7]
- Event-driven, sparse, memory beside compute
- **Dense NPU** (TPU, phone NPU) ≠ **neuromorphic processor**: lower throughput, far lower power and heat
- Chips exist: TrueNorth, Loihi [10, 11]

Device + spikes: **a spiking accelerator for vision that must also be dependable**

[68] Capra 2020 [15] Kim 2007 [7] Roy 2019 [10] Akopyan 2015 [11] Davies 2018 [67] Jouppi 2017

Inside a Time-Multiplexed SNN Accelerator: Who Can Fail?

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ODIN-style accelerator

[52]

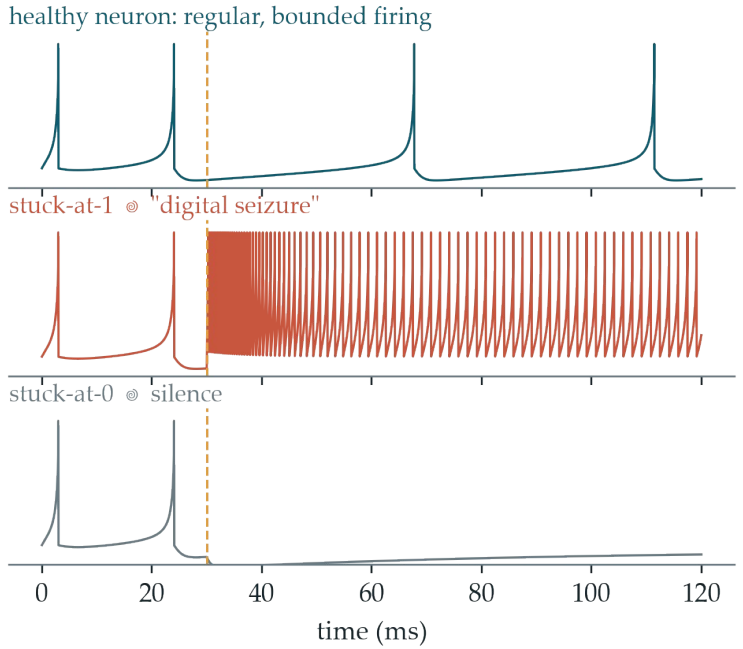
Central controller / FSM

Network-on-chip / AER router

Neuron-state & weight memory

Neuron core ×1…5 (time-multiplexed)

One core serves **hundreds of virtual neurons**: a single permanent defect in it reaches every neuron mapped onto it. Four vulnerable parts — **this thesis hardens the neuron core first**.



SA1 → parasitic current → digital seizure; SA0 → silence [17, 24]

TMR masks it at **3.40× the cells** — the extreme edge cannot pay for triplication

[52] Frenkel 2019 (ODIN) [17] Vatajelu 2019 [24] Spyrou 2021 [16] Hill 2022

Research Gaps

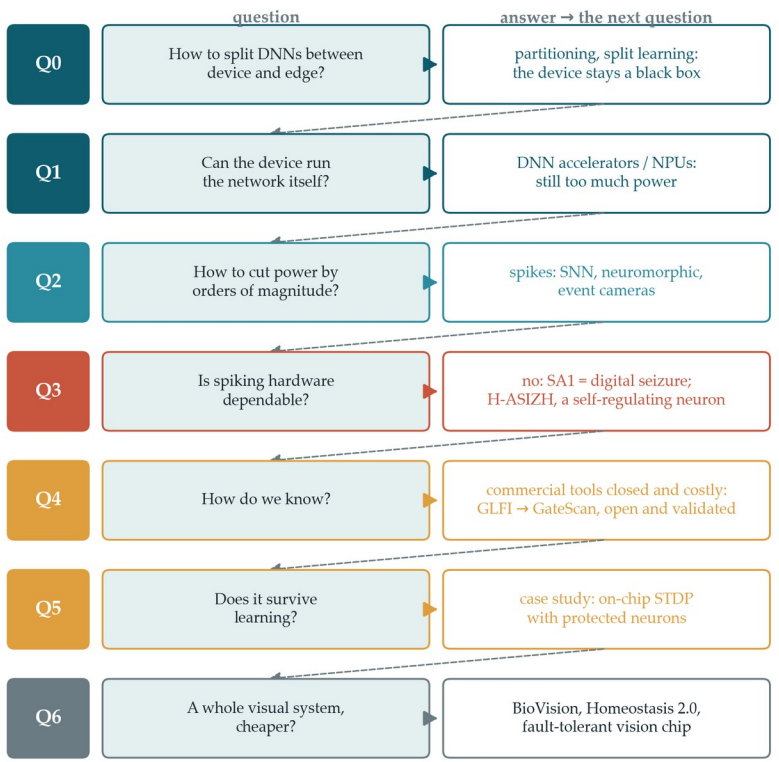
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G1	**The device is an ideal black box**	Edge-intelligence research optimises placement and assumes a correct device [57, 58].
G2	**No dependable path for vision at the extreme edge**	Neuromorphic + event vision is efficient, but its behaviour under permanent faults is little studied [17, 18].
G3	**Fault tolerance does not scale down**	TMR ≈ 3× [19]; SNN schemes need spares, fault maps, re-mapping [22, 34]; supervisors are unprotected [21, 46].
G4	**The neuron is a primitive, not a design object**	Protection acts on, around or above the neuron; no proven bound on its behaviour [24, 47].
G5	**Evaluation is not physically faithful**	Software or two-valued campaigns; faithful gate-level tools are closed and costly [25, 26].
G6	**Dependability and learning are studied apart**	Fault tolerance is tested on frozen networks; compatibility with on-chip learning is rarely shown.

[57] Kang 2017 [58] Zhou 2019 [17] Vatajelu 2019 [18] Schuman 2020 [19] Lyons 1962 [22] ReSpawn [34] RescueSNN [21] SPANNER [46] Johnson 2018 [47] Das 2025 [25] Leveugle 2009 [26] SHADOWFI

Research Questions and Contributions

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C1 — **System argument** — Ch. 1-3

Safety-critical vision must close its loop on the device; fault tolerance moves down the stack.

C2 — **H-ASIZH neuron** — Ch. 4

Exact affine map + homeostatic threshold; Lyapunov proof of a uniformly ultimately bounded rate.

C3 — **Hardware-ready core** — Ch. 5

Powers-of-two DSE, 1-cycle Q7.19 datapath, restoring-organ control plane, P&R on Sky130.

C4 — **Trustworthy fault analysis** — Ch. 6

GLFI: four-valued, in-circuit saboteurs → GateScan: certified vulnerability bracket.

C5 — **Neuron → vision system** — Ch. 7-8

On-chip STDP learning; BioVision-SNN: retina, ventral & dorsal streams, local learning.

02

Literature Review



Accelerators for Vision: Fast, but Assumed Correct

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Work	Class	Mechanism	Fault model	Limitation at the extreme edge
Eyeriss [66]	A · ASIC	Row-stationary dataflow; AlexNet conv 34.7 fps @ 278 mW	—	Assumes correct hardware
TPU [67]	A · ASIC	Systolic array; 15–30× faster than CPU/GPU	—	Data-centre power
UAV detector on FPGA [69]	A · FPGA	Low-precision SSD co-design	—	2.54 W; no protection
Zhang et al. [70]	B · systolic	Fault-aware pruning + retraining	Permanent	Needs fault map and retraining
Li / Ares [71, 72]	B · analysis	Error propagation, resilience quantification	Transient	Analysis only
SalvageDNN [42]	B · mapping	Saliency-driven fault-aware mapping	Permanent	Diagnosis + re-mapping
FORTALESA / reuse [74, 75]	B · systolic	Reconfigurable PE protection; algorithmic reuse	Trans. + perm.	Redundancy inside the PE
Space CNN / YOLO [76, 77]	B · FPGA	TMR / NMR, scrubbing, SET-aware placement	SEU	≈ 3× area and power
None of the above	**—**	**Spiking computation**	**—**	**Not event-driven, not µW-scale**

Protection sits **on, around or above** the processing element — and none of it is spiking

[66] Eyeriss [67] TPU [69] Suh 2023 [70] Zhang 2018 [71] Li 2017 [72] Ares [42] SalvageDNN [29] Libano 2019 [74] FORTALESA [75] Ait Alam 2024 [76] Sterpone 2023 [77] Antunes 2025

Neuromorphic Processors and Neuron Cores

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Chip / core	Year · node	Neurons	Highlight	Fault tolerance
TrueNorth [10, 78]	2014 · 28 nm	1 M, 256 M synapses	≈ 65 mW real time	Not a target
SpiNNaker [79]	2014	ARM many-core	Large-scale real-time simulation	Not a target
Loihi / Loihi 2 [11, 80]	2018 / 2021	Programmable	On-chip learning	Not a target
ODIN [52]	2019 · 28 nm	256 · LIF or 20 Izhikevich behaviours	0.086 mm², 12.7 pJ/SOP, time-multiplexed	Not a target
MorphIC [81]	2019 · 65 nm	4 × 512 LIF	Binary weights, hierarchical routing	Not a target
Speck [82]	2024	327k-neuron CNN + DVS	Clockless; 0.70 mW system	Not a target
QUANTISENC [83]	2024	Configurable LIF	Software-defined fixed-point cores	Not a target
Multiplier-free Izhikevich [13, 84]	2018–2025	Izhikevich	Shift-add, PWL / comb-exp	Not a target

A time-multiplexed core is a single point of failure for hundreds of neurons — **yet no core protects itself**

[10] Akopyan 2015 [78] Merolla 2014 [79] Furber 2014 [11] Davies 2018 [80] Orchard 2021 [52] Frenkel 2019 [81] MorphIC [82] Speck [83] QUANTISENC [13] Haghiri 2018 [84] Sruthi 2025

Fault Tolerance in SNN Hardware, Layer by Layer

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Layer	Representative work	Mechanism	Fault model	Cost / weakness
System & memory	SoftSNN, SparkXD, EnforceSNN [37, 35, 36]	Bound-and-protect; approximate DRAM	Soft / memory	Transient only; relative PPA
Mapping & NoC	ReSpawn, RescueSNN, MigSpike, 3D-NoC [22, 34, 23, 38]	Fault-aware mapping, migration	Permanent	Spares + fault map; timing tied to one die
Bio-inspired supervisor	SPANNER, astrocyte nets, NeuFT [21, 44, 86]	Astrocyte-like monitor beside the neurons	Synapse / SEU	Supervisor on the same die is unprotected
External homeostasis	Johnson et al. [46]	Astrocyte-driven regulation	Permanent	Regulator ≠ neuron
Neuron & structure	Spyrou, dendritic, population coding [24, 47, 88]	Self-test + output TMR; structural; redundant populations	Dead / saturated	Behavioural only; ≈ 60 % overhead; extra neurons
Neuron-intrinsic (this thesis)	**H-ASIZH**	**Adaptive threshold as a state variable + restoring organ**	**Stuck-at, every node**	**1.39× cells (2.31× hardened); proven bound**

Every branch acts on, around or above the neuron — **none acts inside it**

[37] SoftSNN [35] SparkXD [36] EnforceSNN [22] ReSpawn [34] RescueSNN [23] MigSpike [38,40] 3D-NoC [21] SPANNER [43–45] astrocytes [86] NeuFT [46] Johnson [24,87] Spyrou [47] Das [88] Gleyo [90] Stratigopoulos

Fault-Injection and Reliability Tools

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Tool	Injection level	Open	4-valued	Note
Synopsys FuSa / fault simulator	Gate	✗	✓	Licensed, ISO 26262; hard to reproduce outside
SHADOWFI [26]	Netlist (ASIC)	✓	✗	Hyperscale; injection outside the design; Verilog only
RADSAFiE [91]	Netlist (FPGA)	✗	✗	Vendor-locked to Vivado
FIdelity [73]	SW model of accelerator RTL	—	—	46 M injections on NVDLA; transient logic faults
enpheeph / SpikingJET / SpikeFI [92, 93, 94]	Software (SNN)	✓	✗	No hardware fidelity
El-Sayed et al. [89]	Netlist / transistor	✓	✗	Neuron fault models; no automated campaign
GLFI → GateScan (this thesis)	**Gate, in-circuit saboteurs**	✓	✓	**Dual-engine, formal oracle, certified bracket**

Without X/Z and cross-checks, robustness describes **the model, not the circuit** [25]

[26] SHADOWFI [91] RADSAFiE [73] FIdelity [92] enpheeph [93] SpikingJET [94] SpikeFI [89] El-Sayed [25] Leveugle 2009

Bio-Inspired Spiking Vision Systems

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Work	Spiking	Local learning	Streams	Attention	Output
Kheradpisheh 2018 [95]	✓ TTFS	STDP	Ventral	—	Class
Mozafari 2018 [96]	✓ TTFS	R-STDP	Ventral	—	Class
Paredes-Vallés 2019 [98]	✓ events	STDP	Motion	—	Optical flow
StereoSpike 2022 [99]	✓	✗ surrogate	—	—	Dense depth
WWN-2 2010 [100]	✗	Hebbian	What + where	Top-down	Location + identity
SpikeYOLO 2024 [97]	✓ I-LIF	✗ backprop	—	—	Boxes: COCO 66.2 mAP@50
Speck 2024 [82]	✓ + DVS	✗	—	Dynamic	Class at 0.70 mW
BioVision-SNN (this thesis)	**✓ TTFS waves**	**STDP / R-STDP**	**Ventral + dorsal**	**Feedback to LGN**	**What-where, depth, motion, identity**

No spiking model learns **border ownership** locally; feedback attention is shown in biology [102]

[95] Kheradpisheh 2018 [96] Mozafari 2018 [98] Paredes-Vallés 2019 [99] StereoSpike [100] WWN-2 [97] SpikeYOLO [82] Speck [101] Zhou 2000 [102] Debes & Dragoi 2023

Where This Thesis Stands

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Work	Vision	HW (RTL/ASIC)	Spiking	Fault-tolerant	Permanent faults	Formal bound	Gate-level 4-valued	Local learning
Eyeriss, TPU	✓	✓	✗	✗	✗	✗	✗	✗
Zhang 2018, SalvageDNN	✓	✓	✗	✓	✓	✗	✗	✗
Space CNN / YOLO + TMR	✓	◐	✗	✓	◐	✗	✗	✗
ODIN, MorphIC, Speck	◐	✓	✓	✗	✗	✗	✗	◐
SoftSNN, RescueSNN	◐	✓	✓	✓	◐	✗	✗	✗
SPANNER, NeuFT, Johnson	✗	◐	✓	✓	◐	✗	✗	✗
Spyrou 2021, population coding	◐	✗	✓	✓	✓	✗	✗	✗
SpikeYOLO	✓	✗	✓	✗	✗	✗	✗	✗
Kheradpisheh / Mozafari	✓	✗	✓	✗	✗	✗	✗	✓
This thesis	✓	✓	✓	✓	✓	✓	✓	✓

✓ yes ◐ partly ✗ no — references on the previous five slides

03 Methodology

1. **Hardened neuron**
   H-ASIZH + Lyapunov bound
2. **Silicon core**
   1-cycle Q7.19 on SkyWater 130 nm
3. **Control-plane hardening**
   light TMR + restoring organ
4. **Fault analysis**
   GLFI → GateScan
5. **Vision system**
   BioVision-SNN

Methodology: Three Phases, One Device

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done

Phase 1 · Harden the neuron core

- Exact affine map of Izhikevich
- Homeostatic threshold controller
- Lyapunov UUB proof; powers-of-two DSE
- 1-cycle Q7.19 RTL; restoring organ
- P&R on SkyWater 130 nm
- Extended to LIF / IF cores

done

Phase 2 · Trust the measurement

- GLFI: in-circuit saboteurs, 4-valued
- Neuro-metric fault classes
- GateScan: dual-engine (Verilator + Icarus)
- Formal oracle, exact statistics
- Certified vulnerability bracket
- LLM agents propose; the core decides

in progress

Phase 3 · Build the vision system

- BioVision-SNN: retina outside the network
- Ventral + dorsal streams, attention
- TTFS waves, STDP / R-STDP only
- Multiplier-free (shift-add) learning
- Phase 0–1 done; 2–5 to go
- Then: cross-layer fault campaign

Time-multiplexing makes the core the single point of failure → **harden it, prove it, measure it, deploy it**

Phase 1 · The Intrinsic Homeostatic Controller

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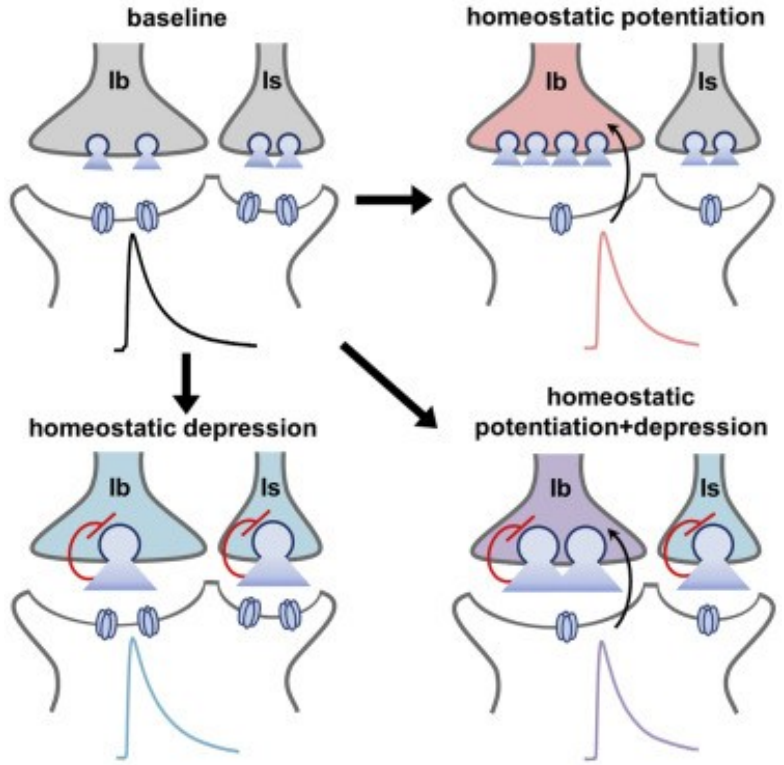
$$d\theta/dt = \alpha\,(\theta_0 - \theta) + \gamma \sum \delta(t - t_k)$$

$$\text{if } V \geq \theta: \quad V \leftarrow c, \quad U \leftarrow U + d + \beta\,(\theta - \theta_0)$$

α — leak
returns θ to θ₀ when the neuron is healthy

γ — penalty
raises θ at every spike

β — brake
deepens the reset of U after a burst



Biological analogue: receptor homeostasis driven by intracellular Ca^{2+} [46]

[46] A. P. Johnson et al., "Homeostatic fault tolerance in spiking neural networks: A dynamic hardware perspective," IEEE Trans. Circuits Syst. I, 2018.

Phase 1 · It Cannot Run Away: Lyapunov Bound

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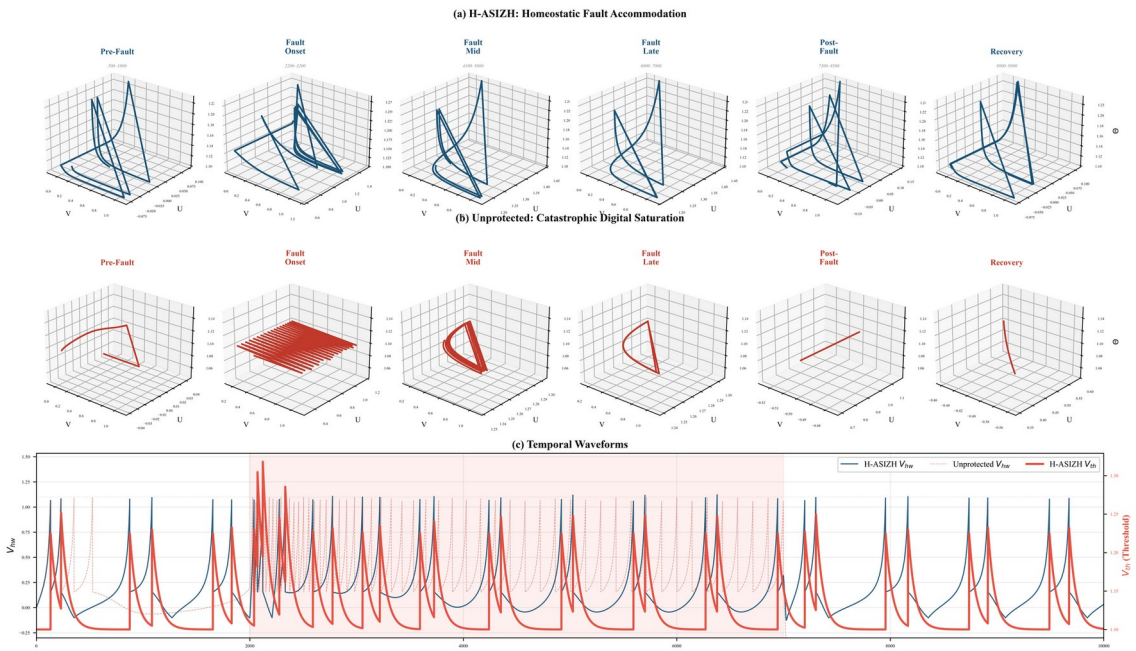
$$\mathscr{V}(\theta) = \tfrac{1}{2}(\theta - \theta^*)^2, \quad \partial r/\partial\theta < 0$$

$$d\mathscr{V}/dt = (\theta - \theta^*)\,[\,-\alpha(\theta - \theta^*) + \gamma\,(r - r^*)\,] < 0$$

Hyperactive phase
$\theta < \theta^* \;\rightarrow\; r > r^* \;\rightarrow\; d\mathscr{V}/dt < 0$

Hypoactive phase
$\theta > \theta^* \;\rightarrow\; r < r^* \;\rightarrow\; d\mathscr{V}/dt < 0$

Uniformly ultimately bounded
band $|\theta - \theta^*| \lesssim \gamma$, set by the penalty, not by the fault [50, 51]



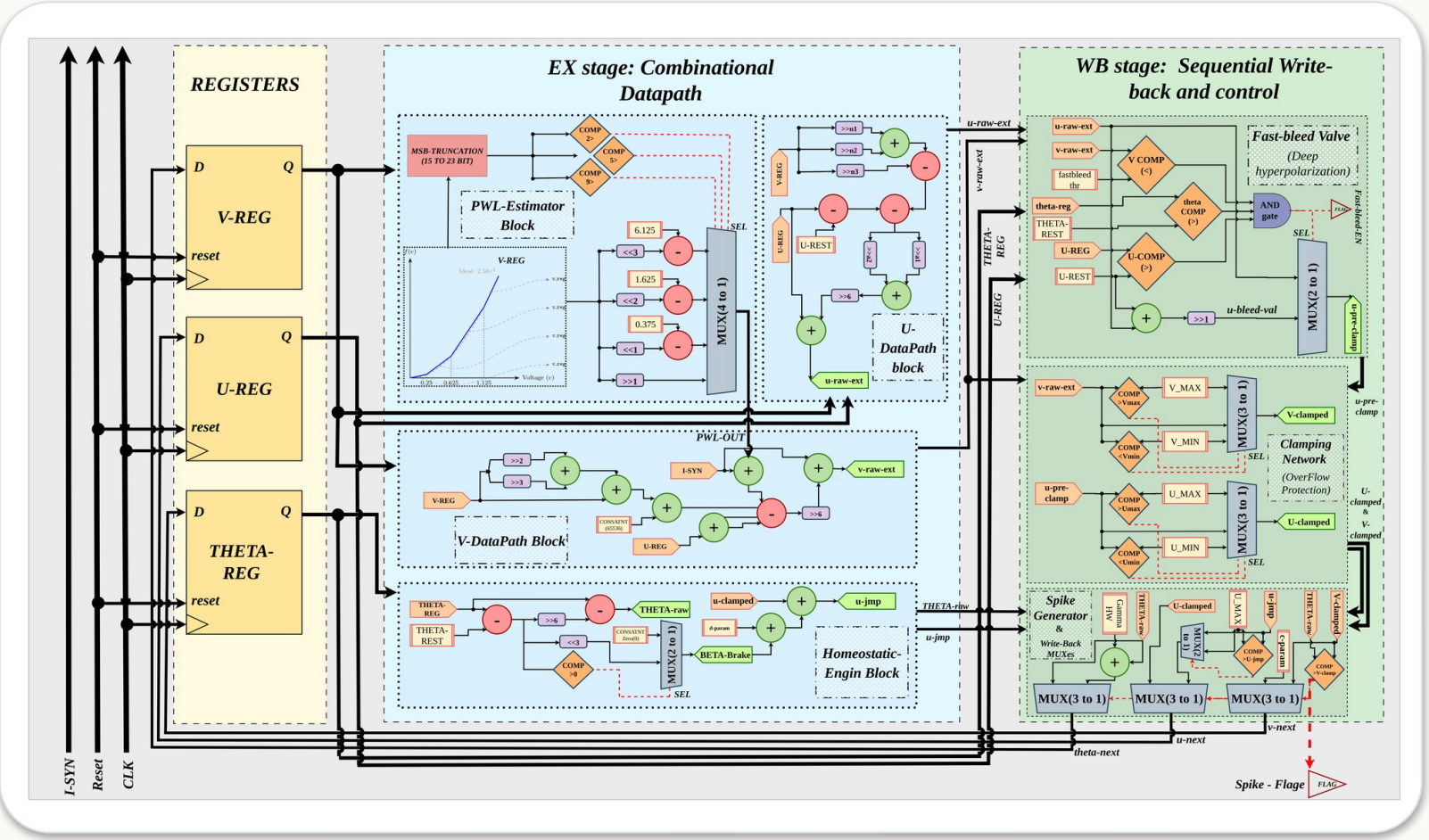
2-D limit cycle → bounded 3-D homeostatic attractor; θ rises and absorbs the defect

[50] H. K. Khalil, Nonlinear Systems, 3rd ed. Prentice Hall, 2002.
[51] J.-J. E. Slotine and W. Li, Applied Nonlinear Control. Prentice Hall, 1991.

Phase 1 · A One-Cycle Micro-Architecture

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EX stage
PWL estimator, V/U datapaths, homeostatic engine

WB stage
clamps, fast-bleed valve, spike generator

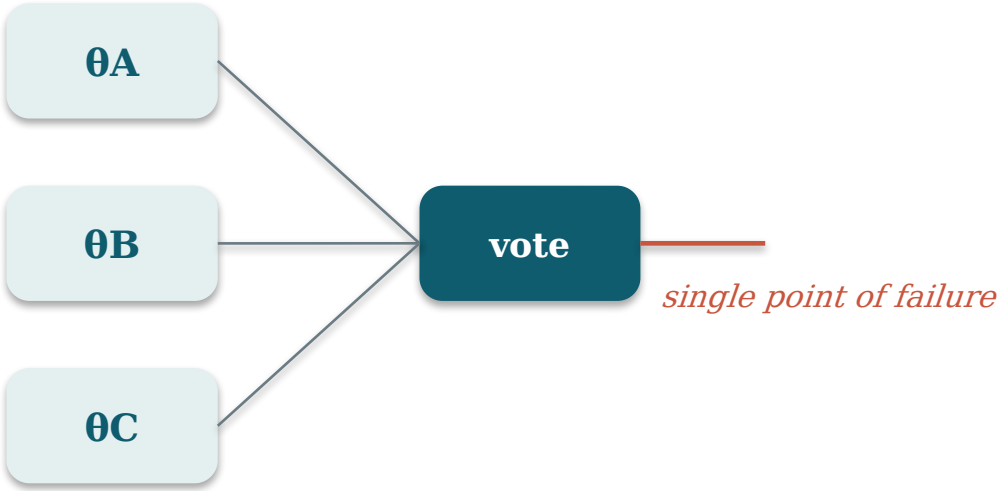
No multiplier
shifts and additions only

1 update / cycle
no latency jitter in a timing code

Phase 1 · Hardening the Control Plane

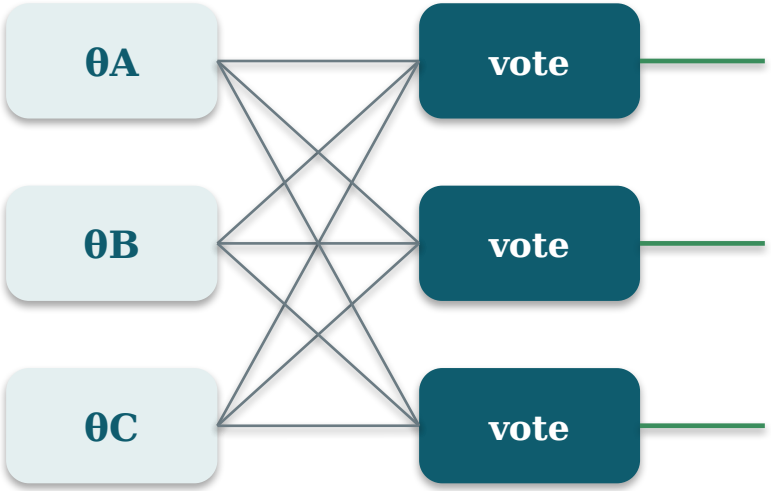
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Light TMR



θ-register + 7 decision flags → +1,500 cells

Restoring organ



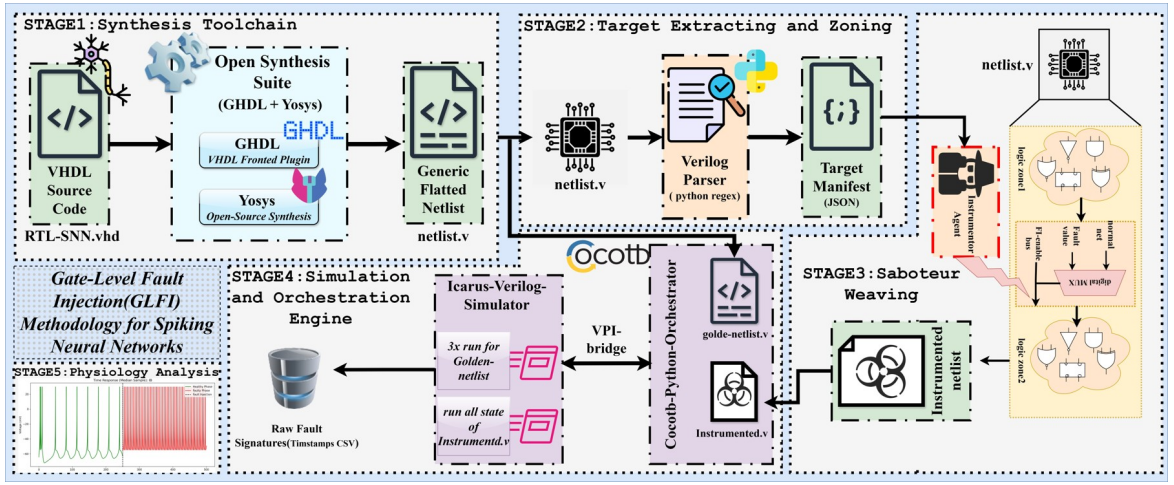
voters triplicated too → +124 cells (≈ 3%)

no single net

V-REG, U-REG, the PWL estimator and both adder trees stay simplex: the controller already protects them.

Phase 2 · From GLFI to GateScan

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Shadow hook, compiled into the netlist: assign net = (fi_enable && fi_target_id==ID) ? fi_value : hook_net;
One instrumented netlist for the whole campaign; fi_value can force 0, 1, X or Z.

2 GateScan Mark III

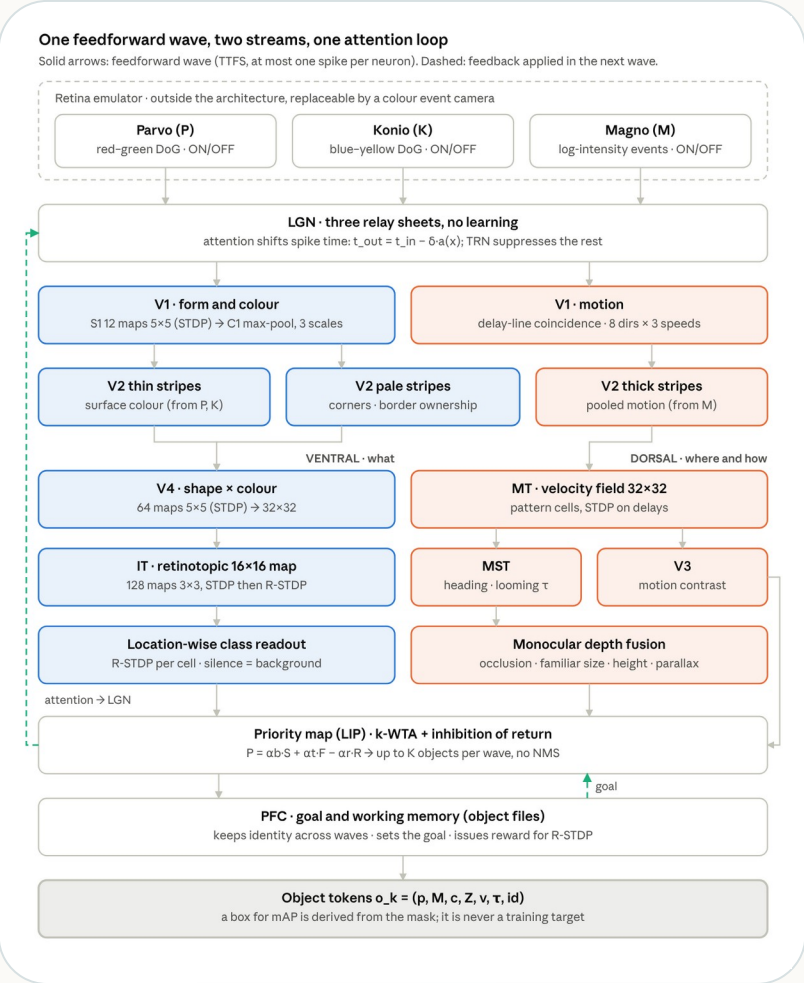
- A certified bracket instead of one SDC %: lower = proven detections, upper = all but proven-harmless
- Exact intervals: Clopper–Pearson, hypergeometric, census
- Escape certificate and an 11-class taxonomy beside SDC/DUE/Masked
- Two engines (Verilator + Icarus/cocotb) and a formal oracle
- LLM agents only propose workloads; a wrong agent can widen the bracket, never falsify it
- **1,834 two-engine runs, 0 disagreements; 13–55 % of "masked" faults proven vulnerable**

GLFI asked "is the neuron robust?" — GateScan asks **"can we trust the answer?"**

[25] Leveugle 2009 [26] SHADOWFI GateScan Mark III.1 (github.com/gatescan-dev/gatescanMarkIII)

Phase 3 · BioVision-SNN Architecture

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Retina (fixed)	P / K / M channels, DoG, divisive normalisation, intensity→latency; microsaccade events (no DVS camera)
LGN	Relay + attention gate: attended spikes fire earlier (δ = 3 steps); virtual fovea
V1 · V2	S1 12 maps (1,800 weights), C1 = earliest spike; thin / pale / thick stripes; direction-selective cells
Ventral	V4 (64) → IT (128); location-wise R-STDP readout: silence = background
Dorsal	MT (aperture problem), V3 motion contrast, MST heading and time-to-contact
Depth · select	5 monocular cues; priority map with k-WTA replaces NMS; PFC object files keep identity
Output	Token o = (position, mask, class, depth, velocity, TTC, id); border ownership B±

TTFS waves: one spike per neuron per wave; worst-case **≈ 46 ms** (wave 1), **≈ 91 ms** (wave 2); ≈ 156k weights

[95] Kheradpisheh 2018 [96] Mozafari 2018 [101] Zhou 2000 [102] Debes & Dragoi 2023 [103] Lichtsteiner 2008 [107] SpykeTorch

Phase 3 · Hardware-Aware Retina and Learning

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1 Log-domain retina

- Binomial filters instead of Gaussians
- Mitchell log: error ≤ 0.0861 octave
- Divisive normalisation = one subtraction
- Brightness invariance ≤ 0.172 octave

2 Multiplier-free learning

- $s(w) = ½ \cdot \min(w, 1-w)$ replaces $w(1-w)$
- Rates = sums of two powers of two
- 16-bit weights; 24–29-bit accumulators
- Accuracy cost ≈ 0 vs. floating point

3 H-ASIZH inside TTFS

- $\theta \leftarrow \theta + \alpha(\theta_0 - \theta) + \gamma \cdot z$ per image
- Calcium $C = (\alpha/\gamma)(\theta - \theta_0) \in [0, 1]$
- β-brake moved into plasticity
- Floor $\theta \geq \theta_0$ silenced S1 → two-sided rate homeostasis (Homeostasis 2.0)

4 Reward learning

- First-spike R-STDP decision layer
- Per-neuron calcium adaptive rate
- AGREL attention rule for hidden layers [106]: +15 % on ETH-80
- SDSP / R-SDSP tested [105]

0 · Reproduce | **1 · Colour ventral** | **2 · Where (readout)** | **3 · Dorsal & waves** | **4 · Attention, depth** | **5 · Borders, faults**

Two design rules for every block: **a biological counterpart** and **shift-add-compare hardware**

[104] Kheradpisheh 2016 [105] Brader 2007 [106] Roelfsema & van Ooyen 2005 [96] Mozafari 2018

04

Results

4.2%

dangerous faults
(12.0% for full TMR)

4.76×

reliability figure of merit
(1.13× for full TMR)

1.47 pJ

per update — the same
energy as full TMR

1020 / 1024

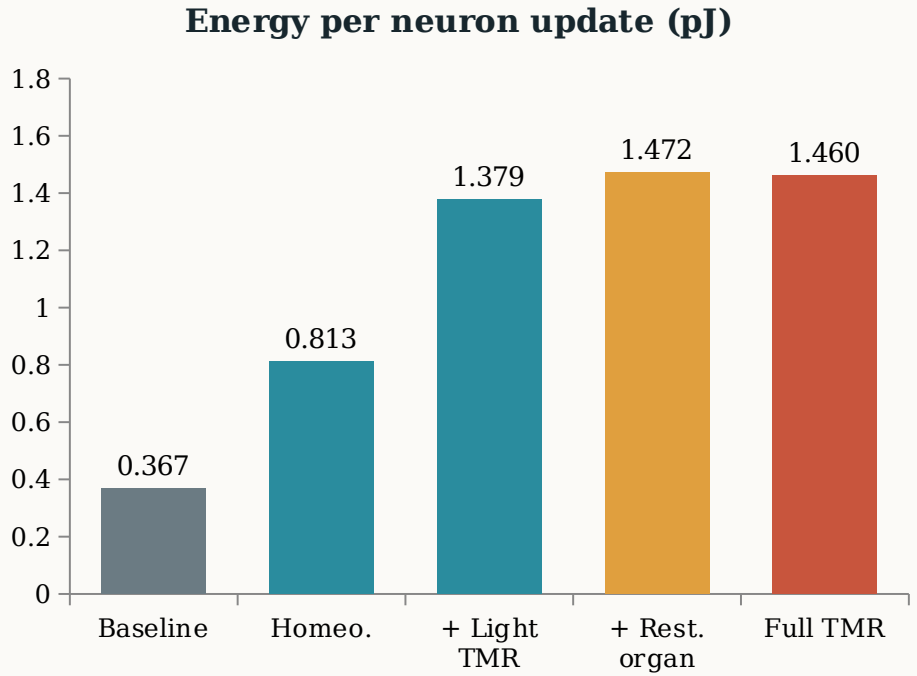
hidden neurons stay
active while learning

Physical Implementation and Energy

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Design	Cells	Fmax (MHz)	Power (µW)	pJ / update
Baseline	1,772	83.61	30.7	0.367
Homeostasis	2,469	64.92	52.8	0.813
+ Light TMR	3,969	64.18	88.5	1.379
+ Restoring organ	4,093	62.11	91.4	**1.472**
Full TMR	6,025	80.00	116.8	**1.460**

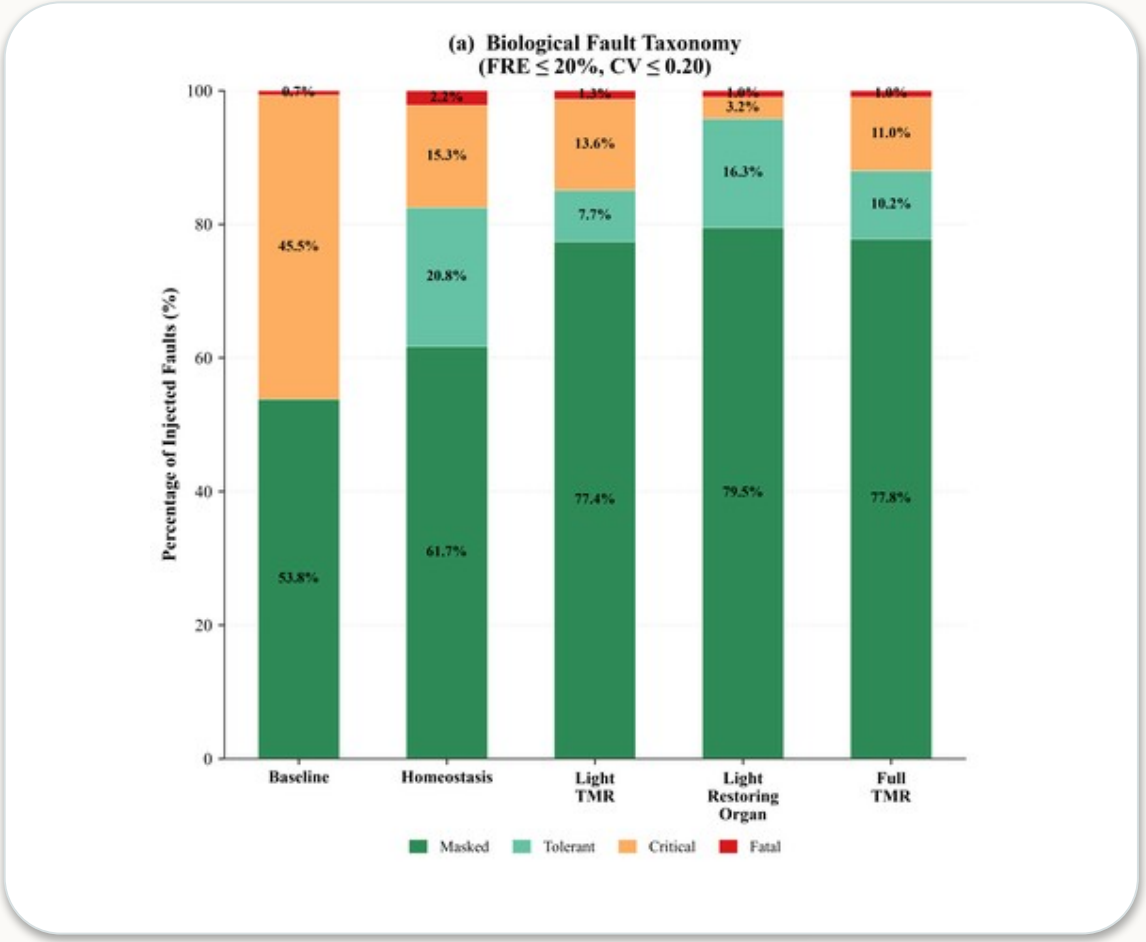
- The controller alone costs 697 cells (1.39×)
- Full TMR is faster but not more efficient



1.472 vs 1.460 pJ
≈ 3× safer at the same energy

SkyWater 130 nm, OpenLane place & route, OpenSTA sign-off at 1.8 V, 25 °C with post-route parasitics; leakage < 0.05% of the total.

Fault Outcome Distribution



Baseline
45.5% critical and no tolerant class: a fault is invisible or destructive

Homeostasis
critical falls threefold to 15.3%; a 20.8% tolerant class appears; fatal rises to 2.2% (brake)

Restoring organ
79.5% masked, 3.2% critical, 1.0% fatal — the floor set by the spike output

Every node injected, both polarities, four-valued logic

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Reliability Figure of Merit (RFoM)

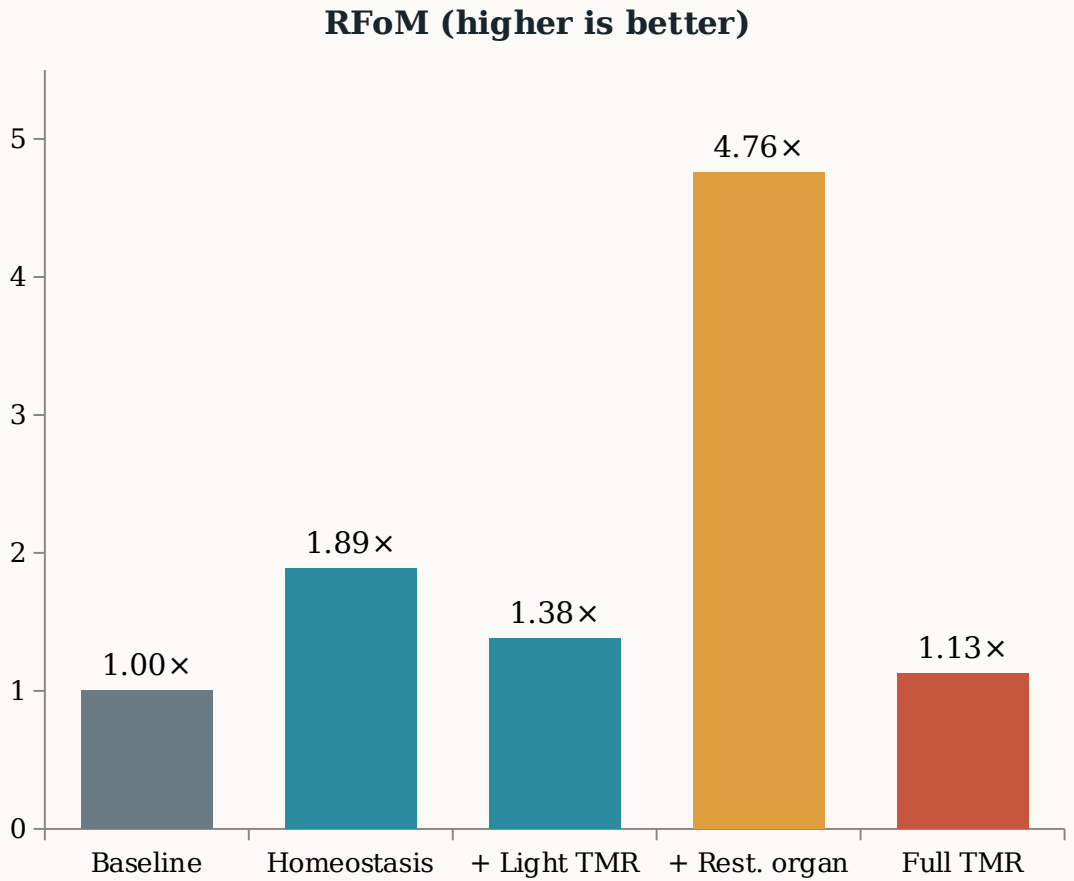
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$$RFoM = (AVF_base / AVF_i) \div (N_i / N_base)$$

vulnerability reduction ÷ area growth

- AVF = (critical + fatal) / all faults
- Worth its silicon only if RFoM > 1
- Full TMR: 3.85× safer but 3.40× larger
- Restoring organ: 11.0× safer, 2.31× larger

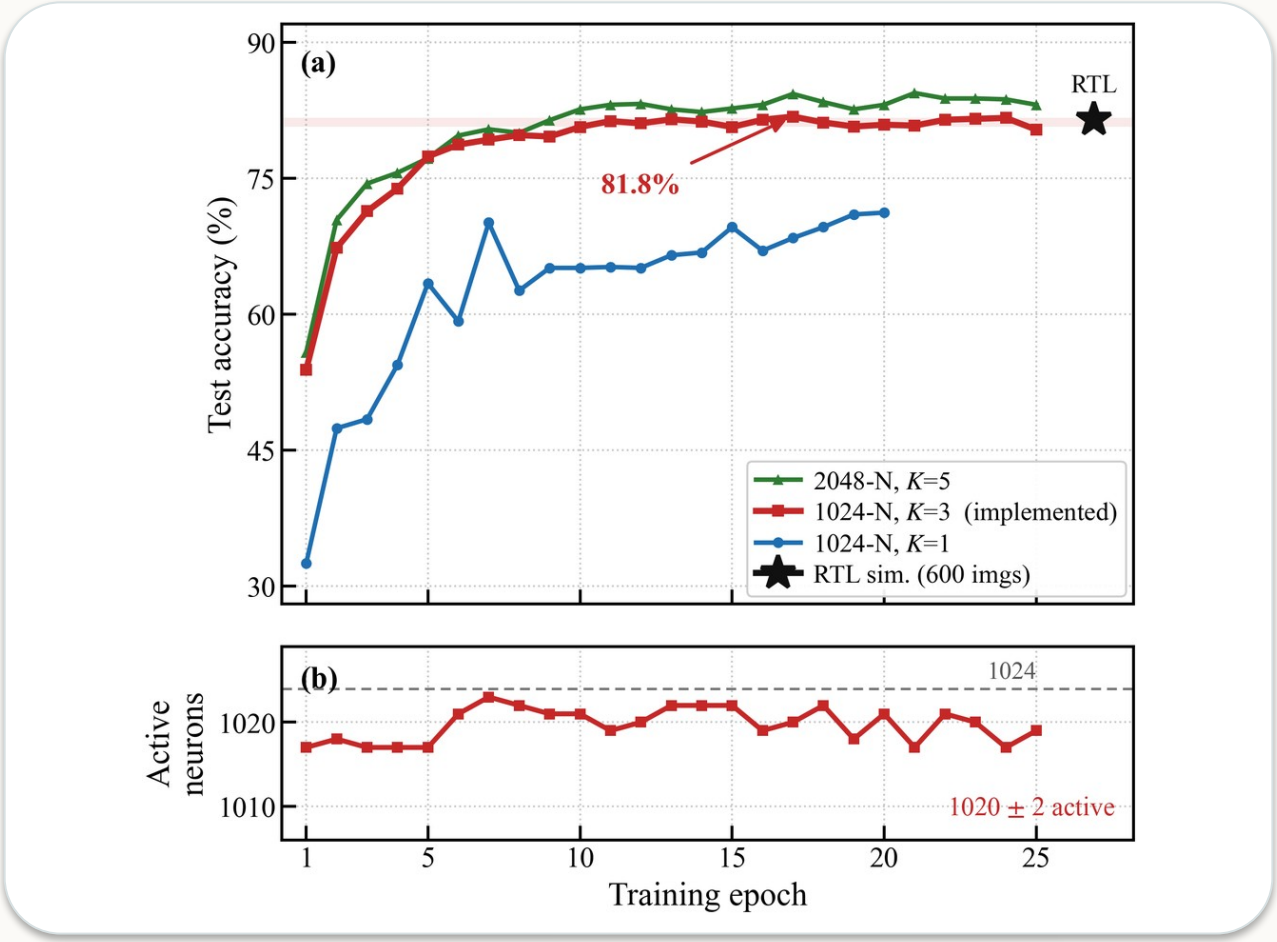
Plain homeostasis already beats full TMR by 1.67×



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Phase 1 · On-Chip Learning Case Study

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81.8%
peak; 81.2 ± 0.5% steady state

81.5%
synthesisable RTL, 600 test images

1020 ± 2 / 1024
hidden neurons stay active

The same θ that stops a runaway also stops neuron death and domination.

BioVision-SNN: Results So Far (Phases 0–1)

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Phase 0 (test %)	Paper	Ours
RL · Caltech (Mozafari)	98.9 ± 0.4	98.65 ± 0.49
RL · ETH-80 leave-one-out	89.5 ± 1.9	81.83 ± 5.58
Probe · Caltech (SDNN)	99.1 ± 0.2	98.96 ± 0.25
Probe · MNIST (SDNN)	98.4	98.52 ± 0.06
Probe · ETH-80 5/5 (SDNN)	82.8	84.25 ± 3.49
Multiplier-free mode	—	−0.9 / +1.7 pp

H-ASIZH threshold gave the best ETH-80 features (84.25 %); per-neuron calcium rate matched Mozafari on Caltech; decisions with < 1,000 input spikes.

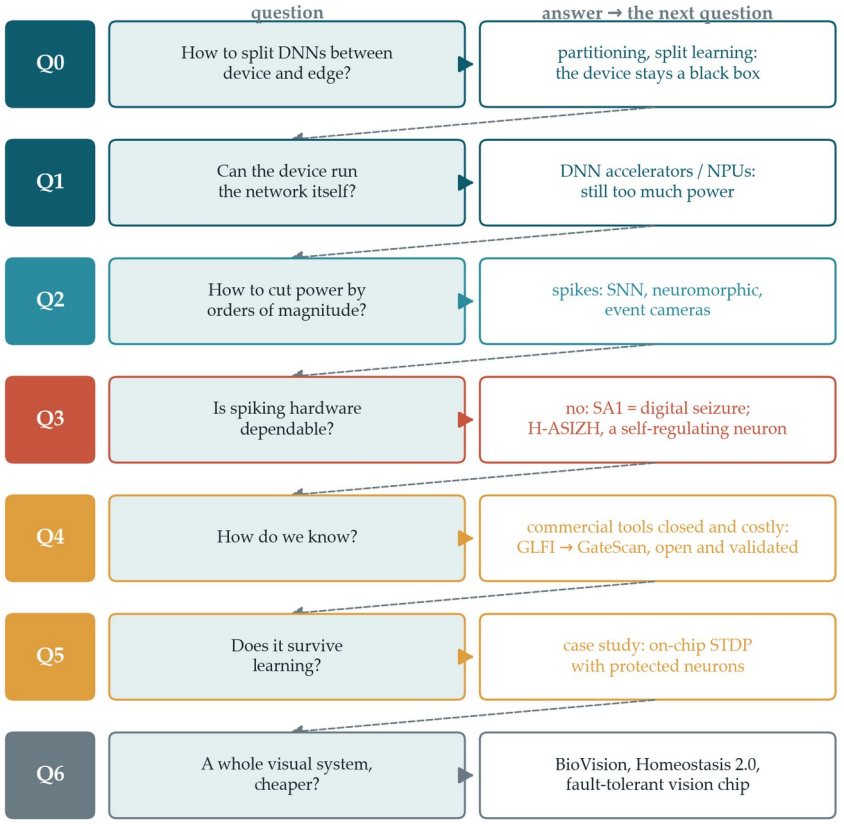
1 Phase 1 (draft, 128 px)

- Colour helps: P+K +6.7 pp over grey (RL, ETH-80 val.)
- RL ETH-80 test: 72.87 ± 2.22 (decision still rising at epoch 80)
- Caltech-4, grey retina: 89.42 ± 1.18
- Darker images: invariant (log-domain theorem)
- Gamma 0.5 collapsed accuracy → contrast adaptation (running)
- Found and fixed: K-channel background artefact, power-of-two rate collapse, 3 errors in the design maths (107 unit tests)

Local learning reproduces — and on MNIST and ETH-80 exceeds — the SDNN baseline, **multiplier-free**

[95] Kheradpisheh 2018 (SDNN) [96] Mozafari 2018 — all numbers: test accuracy, mean ± std over seeds / folds; Phase 0 gate closed 2026-10-20

05 Conclusion



Conclusion

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A Dependability is a device requirement

For safety-critical vision the loop must close on the device; fault tolerance has to move down the stack with the computation.

B Fault tolerance can live inside the neuron

H-ASIZH: proven bounded rate; 4.2 % vs 12.0 % dangerous faults, RFoM 4.76× vs 1.13×, at the same 1.47 pJ per update.

C The measurement itself must be trusted

GLFI → GateScan: four-valued, in-circuit, cross-checked; a certified bracket instead of an optimistic point estimate.

D From a neuron to a vision system

On-chip learning keeps 1020/1024 neurons alive; BioVision reproduces the SDNN baselines with local, multiplier-free learning.

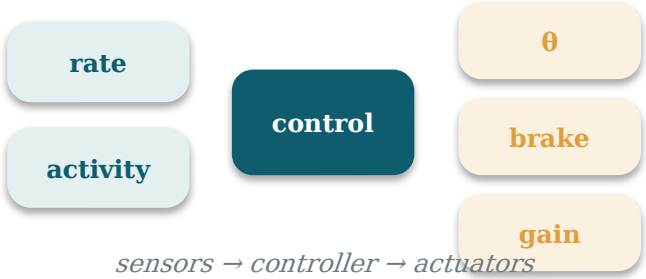
Triplication buys nothing that **accommodation** does not buy more cheaply

Future Work: Where This Goes Next

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in progress

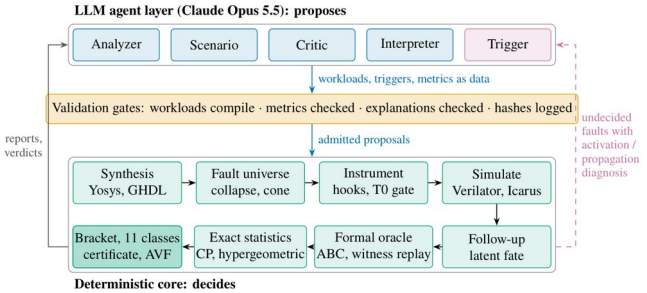
Homeostasis 2.0 on cheaper cores



- From Izhikevich to LIF / IF cores
- Replace TMR and the restoring organ with a richer loop: 2 sensors, 3 actuators
- Different mechanisms, one goal — degeneracy, as in biology [53]

in progress

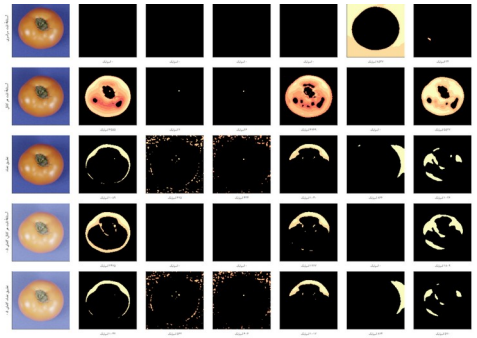
GLFI → GateScan



- An open fault-analysis application: RTL in, per-node vulnerability out
- LLM agents propose workloads; simulation and formal proof decide
- 0 disagreements in 1,834 cross-checks; agent workloads lift ITC'99 b13 detection 35.7% → 86.0%

in progress

BioVision-SNN on fault-tolerant cores



- **Phases 0–1 done: retina emulator and colour ventral stream; next: location-wise readout, dorsal stream, attention, depth, border ownership**
- **Cross-layer campaign: gate-level fault outcomes injected per layer of the vision system**

[53] E. Marder and J.-M. Goaillard, "Variability, compensation and homeostasis in neuron and network function," Nat. Rev. Neurosci., vol. 7, pp. 563–574, 2006.

Research Articles and Artefacts

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1. **H-ASIZH: A Highly Reliable Digital Izhikevich Neuron Core Using Adaptive Threshold Control for Fault Accommodation**
   A. Roshani, M. Ansari, M. T. Manzuri Shalmani
   IEEE TCAS-I · submitted

2. **GateScan: certified gate-level vulnerability analysis (working title “Masked Is Not Harmless…”)**
   GateScan Mark III.1
   IEEE TCAD · in preparation

3. **BioVision-SNN: a hardware-aware bio-inspired spiking vision system with local learning**
   Phase reports 0–1
   in preparation

Open source: github.com/aliroshanidino/GLFI_fault_campaign · github.com/gatescan-dev/gatescanMarkIII (MIT)

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Thank you

Questions and discussion

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